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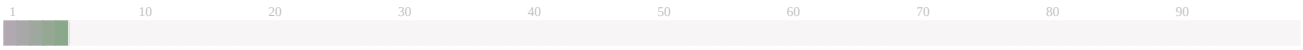
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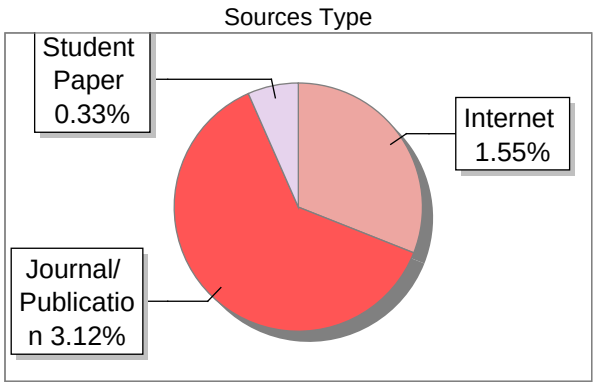
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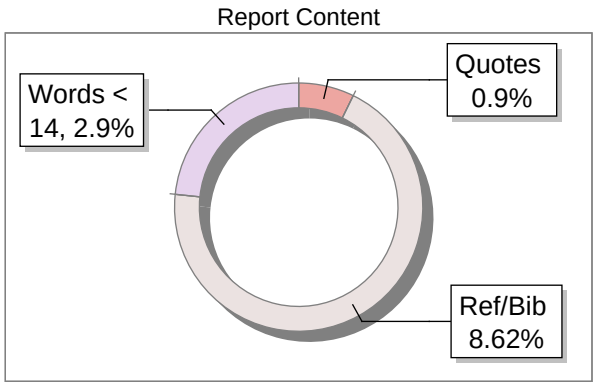
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VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Belagavi, Karnataka State, India

A project report on

"Threat/Vulnerability Automation Detection System"

Submitted in partial fulfillment of the degree of

Master of Computer Applications

Submitted by

Ms. Shashikala Karabasappa Mokashi

2BA23MC044

Under the Guidance of

Prof. M. H. Shirur

Dept. of MCA



DEPARTMENT OF MASTER OF COMPUTER APPLICATIONS (MCA)

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Visvesvaraya Technological University

Belagavi, Karnataka State, India.



Department of Master of Computer Applications (MCA)

Basaveshwar Engineering College

Bagalkote – 587 102

CERTIFICATE

This is to certify that **Ms. Shashikala Karabasappa Mokashi,** bearing USN **2BA23MC044,** has satisfactorily completed the project work entitled **'Threat/Vulnerability Automation Detection System'** submitted to the Basaveshwar Engineering College, Bagalkote, in partial fulfillment of the requirements for the award of Master of Computer Applications (MCA), during the academic year 2024-2025.

M. H. Shirur	**Mr. Sadhan**
Internal Guide	**External Guide**

Prof. M. H. Shirur	**Prof. Sudha K. S.**	**Dr. B. R. Hiremath**
Project Coordinator	**Head of the Department**	**Principal**

Date of Submission:

Examiners:

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Declaration

I, **Shashikala Karabasappa Mokashi**, student of final semester MCA, Basaveshwar Engineering College, Bagalkote, hereby declare that the dissertation entitled "**Threat/Vulnerability Automation Detection System**" has independently carried out by me at Master of Computer Application (MCA) Department, Basaveshwar Engineering College, Bagalkote, and submitted in the partial fulfillment of requirements for the award of MASTER OF COMPUTER APPLICATIONS by Basaveshwar Engineering College, Bagalkote, during the academic year of 2024-2025. Further the matter embodied in dissertation has not been submitted by anybody for the award of any degree or diploma to any other university.

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Date: **2BA23MC044**

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Yours Sincerely,
Shashikala Karabasappa Mokashi
2BA23MC044

Abstract

Case Investigation Management System is a safe, central, and web-based system intended for the digitalization of the entire criminal investigation process. Manual documentation and unorganized computer tools in the conventional criminal investigation processes may generate issues of data accuracy, efficiency, and evidence authenticity or safety, among other concerns. The Crime Investigation Management System aims to provide an organized system incorporating the management of FIRs, case follow-ups, evidence, tasks, court trials, and audits.

The software was designed with Python programming with Flask, together with the SQL Alchemy Object-Relational Mapping tool, which was supported by a relational database design. The main functionality of CIMS, however, lies in its robust evidence management system, which uses SHA256 cryptographic hashing with a sound chain-of-custody approach to verify evidence integrity and admissibility. Finally, the software incorporates other functionalities such as Role-Based Access Control, authentication, as well as logging mechanisms that improve confidentiality, transparency, and accountability.

Experimental testing has proved that this leads to a substantial improvement in the effectiveness of the investigations and a reduced workload. The project is a valuable implementation of secure software principles. It has a high academic quality as the principles of the design of secure software were correctly taken into consideration. The system is a good foundation for the future of modern policing. It will be possible in the future to integrate additional features into this project.

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1. Introduction

Crime investigation is a structured and legally governed process undertaken by law enforcement agencies to identify offenders, collect and preserve evidence, reconstruct events, and present findings before judicial authorities. With increasing crime rates, digital evidence proliferation, and complex legal requirements, traditional paper-based investigation systems have become inefficient and error-prone. Manual handling of FIRs, case diaries, evidence registers, and court documents often results in delayed investigations, misplaced records, lack of transparency, and difficulty in accountability.

The integration of information technology into policing has led to the evolution of digital crime management and investigation systems. A Case Investigation Management System (CIMS) is a centralized digital platform that automates and integrates all investigation-related activities, including FIR registration, case assignment, evidence handling, task management, documentation, and court proceedings. By digitizing the entire investigation lifecycle, CIMS improves operational efficiency, ensures legal compliance, enhances data security, and supports informed decision-making.

The proposed CIMS is a web-based application designed to support multiple stakeholders such as administrators, police officers, investigators, evidence technicians, analysts, and court liaison officers. The system emphasizes secure access, role-based permissions, evidence integrity, audit trails, and standardized investigation workflows, making it suitable for real-world law enforcement as well as academic and training environments.

2. Literature Survey

The literature survey examines existing research, systems, and technologies related to crime investigation management and law enforcement information systems. Studies highlight that digitization significantly enhances investigation efficiency, data accuracy, inter-departmental coordination, and transparency. Modern policing increasingly relies on Records Management Systems (RMS), Crime Information Systems (CIMS), Digital Evidence Management Systems (DEMS), and Case Tracking Systems.

Research also emphasizes the importance of maintaining chain-of-custody for evidence, secure access control, audit logging, and compliance with legal standards. While several systems address isolated aspects of investigation management, comprehensive platforms integrating FIRs, cases, evidence, tasks, and court proceedings remain limited, especially in developing regions.

2.1 Existing System

In the existing system, the management of criminal investigations is predominantly handled through manual and semi-digital processes. Most law enforcement agencies rely on paper-based documentation, physical registers, and isolated computer systems to record and manage investigation-related data. The First Information Report, which marks the beginning of any criminal investigation, is typically recorded manually in registers or as unstructured digital documents without standardized formats or centralized storage.

Case files in the existing system are maintained as physical folders containing handwritten notes, printed reports, photographs, and legal documents. Each investigation officer maintains separate case diaries, making it difficult to consolidate information or track real-time case progress. Evidence records are logged manually using evidence registers, where details such as evidence description, date of seizure, and storage location are written by hand. This approach lacks verification mechanisms to ensure the integrity and authenticity of evidence over time.

In many departments, digital tools such as spreadsheets or basic databases are used to store crime data. However, these tools operate in isolation and are not interconnected.

There is no unified platform to link initial records, accused persons, victims, witnesses, evidence, tasks, and court proceedings. As a result, investigators must refer to multiple sources to obtain a complete view of a case, which increases workload and the possibility of human error.

User access control in the existing system is minimal or nonexistent. Sensitive legal forum data can often be accessed or modified without proper authorization, increasing the risk of data misuse or tampering. There is also no role-based access mechanism, meaning that administrative users, officers, and support staff may all have similar levels of access, which violates security and confidentiality principles.

Another major limitation of the existing system is the absence of audit trails and system activity records. Actions such as case updates, evidence handling, or document modifications are not systematically recorded. This makes it difficult to establish accountability or trace responsibility in the event of discrepancies or legal disputes. Additionally, manual systems do not support automated notifications or alerts, causing delays in task execution, hearings, and investigation milestones.

Court-related processes in the existing system are also fragmented. Court dates, legal sections, judgments, and hearing outcomes are recorded separately, often on paper, making it difficult to track judicial progress or retrieve historical court data efficiently. Report generation for higher authorities requires manual compilation of information, which is time-consuming and error-prone.

Overall, the existing system lacks integration, security, scalability, and efficiency. The dependence on manual processes results in delayed analysis, increased administrative burden, poor coordination among departments, and challenges in maintaining legal compliance. These limitations highlight the urgent need for a centralized, secure, and automated system like the proposed Case Investigation Management System .

2.2 Problems in Existing System

The existing system used for managing criminal probes suffers from several critical limitations due to its heavy reliance on manual processes and disconnected digital tools. One of the most significant problems is the lack of centralized data management.

Investigation-related information such as FIRs, case details, data records, witness statements, and court documents are stored across multiple physical registers and isolated digital files. This fragmentation makes it extremely difficult for investigators to obtain a complete and accurate overview of a case at any given time.

Another major issue is the risk of data loss, damage, or tampering. Physical documents are vulnerable to loss due to mishandling, fire, water damage, or unauthorized access.

Since evidence details are often recorded manually without cryptographic verification, there is no reliable way to ensure the integrity of evidence throughout the investigation lifecycle. This poses serious legal challenges, as compromised evidence can weaken cases in court or lead to acquittals.

The existing system also lacks proper access control and security mechanisms. Sensitive case data can often be accessed or modified by unauthorized personnel due to the absence of role-based access control. Without defined user roles such as investigator, evidence technician, or court liaison, confidentiality and data privacy are compromised. This not only increases the risk of data misuse but also violates standard cybersecurity and legal compliance requirements.

Inefficient case tracking is another prominent problem. In the current system, there is no structured method to track investigation stages or monitor progress in real time. Officers must manually update proceedings diaries and registers, which makes it difficult for supervisors to assess investigation status, identify delays, or reassign resources when required. This often results in prolonged investigations and missed deadlines.

The manual handling of tasks and schedules further reduces operational efficiency. Activities such as evidence collection, witness examination, forensic analysis, and court hearings are tracked informally or through handwritten notes. The absence of automated task management and reminders leads to missed follow-ups, delayed hearings, and poor coordination among investigation teams.

The existing system also fails to provide audit trails and accountability. Actions such as affair updates, evidence transfers, or document modifications are not logged systematically. In the event of discrepancies, it becomes difficult to determine who performed a particular action and when. This lack of traceability weakens internal accountability and creates challenges during audits or legal scrutiny.

Report generation in the existing system is time-consuming and error-prone. Senior officers and judicial authorities often require periodic reports on case status, crime trends, summaries. Since data is scattered across multiple sources, reports must be compiled manually, increasing the likelihood of inconsistencies and inaccuracies.

Additionally, the system does not support scalability or future expansion. As the volume of cases and digital evidence increases, manual systems struggle to handle the growing workload. The absence of analytics, automated insights, and data-driven decision support further limits the effectiveness of investigation processes.

In summary, the existing system is characterized by poor integration, weak security, lack of transparency, and operational inefficiency. These problems significantly hinder effective investigation, delay justice delivery, and increase administrative burden. The identified limitations clearly justify the need for a centralized, secure, and automated solution such as the proposed Management System .

2.3 Available solution and their feature

Several digital solutions have been developed to support law enforcement agencies in managing crime-related data and investigation activities. These solutions aim to reduce manual workload, improve data accessibility, and enhance operational efficiency. However, most existing solutions focus on specific aspects of investigation management rather than providing a comprehensive end-to-end platform.

One commonly available solution is the Records Management System. RMS platforms are designed to store and manage crime records, FIR details, and basic case information in a digital format. They provide functionalities such as data entry, search, and report generation. While RMS solutions help in digitizing crime data, they often lack advanced features such as investigation stage tracking, evidence integrity verification, and real-time task management.

Another available solution is the Crime Information Management System , which focuses on maintaining centralized crime databases for analysis and intelligence purposes. CIMS platforms allow agencies to analyze crime patterns, identify repeat offenders, and generate statistical reports. Although these systems are useful for strategic decision-making, they generally do not support day-to-day investigation workflows, evidence handling, or court process tracking.

Digital Evidence Management Systems are also widely used to store and manage digital evidence such as images, videos, and audio recordings. These systems provide secure storage, access control, and basic chain-of-custody records. However, solutions typically operate as standalone platforms and are not integrated with FIR registration, case management, or judicial processes. As a result, investigators must use multiple systems to manage a single case.

Some agencies adopt Task and Workflow Management Tools to assign duties and monitor investigation activities. These tools offer task scheduling, reminders, and progress tracking. While they improve coordination, they are not specifically designed for criminal investigations and lack features such as evidence linking, legal documentation, and case lifecycle management.

Commercial and proprietary law enforcement software packages are also available, offering a combination of crime records, case tracking, and reporting features. However, these systems are often expensive, require specialized infrastructure, and may not be customizable to local legal procedures or institutional requirements. Additionally, licensing costs and vendor dependency limit their adoption, especially in academic or small-scale environments.

A major limitation across most available solutions is the absence of complete integration. Data records, cases, evidence, tasks, documents, and court proceedings are managed in separate modules or separate systems, leading to fragmented data and inefficiencies. Furthermore, many existing solutions do not emphasize role-based access control, detailed audit logs, or cryptographic evidence integrity mechanisms, which are critical for secure and legally compliant investigations.

In contrast to these partial solutions, there is a clear need for an integrated system that combines all investigation-related functionalities into a single, secure platform. The proposed Case System is designed to overcome the limitations of existing solutions by providing unified case management, secure evidence handling with integrity checks, role-based access control, activity logging, and court process integration within a web-based architecture.

2.4 Proposed Solution (Generic)

The proposed solution is the design and development of a centralized, secure, and web-based Case Investigation Management System that addresses the limitations of existing manual and fragmented digital systems.

The system is intended to provide an integrated platform that supports the complete lifecycle of criminal investigations, from FIR registration to case closure and court proceedings.

The proposed system aims to digitize and automate core investigation processes by consolidating FIR management, case tracking, proof handling, task management, documentation, and judicial workflows into a single unified application.

By adopting a centralized database architecture, the system ensures that all investigation-related data is stored consistently and can be accessed in real time by authorized users.

A key aspect of the proposed solution is the implementation of role-based access control (RBAC). Different stakeholders such as administrators, officers, investigators, evidence technicians, analysts, and court liaison officers are assigned specific roles and permissions. This ensures that sensitive information is accessed only by authorized personnel, thereby enhancing confidentiality, data security, and accountability.

The proposed system also emphasizes secure evidence management and integrity assurance. Evidence items are digitally recorded, categorized, and linked to their respective cases. Integrity verification mechanisms, such as cryptographic hashing, are used to ensure that evidence remains untampered throughout the investigation process. Additionally, a structured chain-of-custody mechanism records every action performed on evidence, providing legal reliability and traceability.

To improve operational efficiency, the proposed solution incorporates task and workflow management features. Investigation activities such as evidence collection, witness examination, forensic analysis, and court hearings can be scheduled, assigned, and monitored within the system. Automated notifications and alerts help ensure timely execution of tasks and reduce delays in investigation processes.

The system also supports document and court case management, allowing users to upload, store, retrieve, and manage legal documents such as charge sheets, forensic reports, and court orders. Court-related information including hearing dates, applied legal sections, judgments, and trial status can be tracked alongside case data, ensuring better coordination between investigative and judicial processes.

From a technical perspective, the proposed solution adopts a modular and scalable web-based architecture. Using widely accepted technologies such as Python and the Flask framework, the system ensures flexibility, maintainability, and ease of future enhancement. The use of a relational database supports data consistency, while secure authentication mechanisms protect system access.

Overall, the proposed generic solution focuses on integration, security, transparency, and efficiency.

By replacing manual and disconnected systems with a unified digital platform, the Case Investigation Management System provides a robust foundation for modern, legally compliant, and technology-driven criminal investigations.

2.5 Advantages of Proposed System

The proposed one offers several significant advantages over traditional manual systems and existing fragmented digital solutions. By integrating all inquiry-related activities into a single, centralized platform, the system enhances operational efficiency, security, transparency, and accountability across the entire investigation lifecycle.

One of the primary advantages of the proposed solution is centralized case management. All information related to first records, cases, accused persons, victims, witnesses, testimony, tasks, documents, and court proceedings is stored in a unified database. This eliminates data fragmentation and enables investigators and supervisors to access complete and up-to-date case information in real time, leading to faster and more informed decision-making.

The system provides enhanced security and controlled access through role-based access control (RBAC). By assigning specific permissions to different user roles such as administrators, investigators, evidence technicians, and court liaison officers, the system ensures that sensitive information is accessible only to authorized personnel. This significantly reduces the risk of data misuse, unauthorized modifications, and confidentiality breaches.

Another major advantage is secure evidence handling and integrity assurance. The proposed solution incorporates digital evidence management with integrity verification mechanisms such as cryptographic hashing. Each piece of evidence is tracked through a structured chain-of-custody process, ensuring that every interaction with the evidence is recorded.

This strengthens the legal credibility of evidence and reduces the possibility of disputes during court proceedings.

The draft system also improves investigation efficiency and workflow management. Built-in task assignment, scheduling, and notification features help investigators manage their activities more effectively. Automated reminders reduce the likelihood of missed deadlines, delayed hearings, or incomplete investigation steps, thereby accelerating case resolution.

Improved transparency and accountability is another key benefit. The system maintains detailed activity logs and audit trails that record all user actions, including case updates, evidence transfers, and document modifications.

This ensures traceability, facilitates internal audits, and strengthens administrative oversight.

The planned solution simplifies report generation and data retrieval. Since all information is stored digitally and structured systematically, reports related to fir, case status, evidence summaries, and court proceedings can be generated quickly and accurately. This reduces manual effort and minimizes the risk of errors.

From a scalability perspective, the system is designed to support future growth and enhancement. Its modular web-based architecture allows for easy integration of advanced features such as analytics, biometric authentication, mobile access, and AI-based crime pattern analysis. This makes the solution adaptable to evolving law enforcement requirements.

Additionally, the conceptual system reduces administrative workload and paperwork, allowing officers to focus more on investigative tasks rather than manual record-keeping. By improving coordination between departments and stakeholders, the system contributes to faster investigations and more effective justice delivery.

In summary, the Case Investigation Management System provides a comprehensive, secure, and efficient digital solution that overcomes the limitations of existing systems. Its advantages directly align with the objectives of modern public safty department by enhancing accuracy, accountability, transparency, and operational effectiveness.

3. Problem Definition

3.1 Problem Definition

The core problem addressed by this project is the lack of a unified, secure, and verifiable digital infrastructure to manage the complex lifecycle of a criminal investigation. Conventional methods of managing police records often suffer from the following critical deficiencies which this project aims to resolve:

- **Fragmented Data Silos:** In traditional systems, the First Information Report , case files, criminal records, and evidence logs are often maintained as disconnected entities. This makes it difficult to link an initial one to a specific judicial Case (represented by the Case model) and its associated suspects (the Criminal model).
- **Lack of Evidence Integrity:** There is no automated mechanism in manual systems to guarantee that digital or physical evidence has not been altered during the investigation. The system must address this by implementing cryptographic verification (SHA256 hashing) for every piece of evidence stored in the Evidence model.
- **Broken Chain of Custody:** Tracking who handled a piece of evidence, when, and where (from collection to court) is prone to human error. The project defines the need for a rigorous digital trail, implemented via the Evidence Chain model, to record every transfer and action taken on an item.
- **Inconsistent Investigation Workflows:** Investigations often lack a standardized progression, leading to delays or missed procedural steps. The system must define and enforce a linear progression of stages—from "Preliminary Enquiry" to "Charge Sheet"—through the Investigation Stage model to ensure procedural compliance.
- **Coordination Gaps in Trial Management:** Coordination between investigative officers and the judiciary is often delayed. The system must provide a bridge by linking investigative cases to the Court Case and Hearing models, ensuring that trial schedules and judicial orders are instantly accessible to relevant officers.

- **Role-Based Security Risks:** Sensitive criminal data requires strict access controls to prevent unauthorized tampering. The problem is defined as the need for a multi-layered permission system where different users (defined by ROLES such as Admin, Investigator, and Evidence Technician) can only perform actions relevant to their specific duties. In addition to the above limitations, several deeper system-level and operational challenges further justify the need for a comprehensive digital investigation framework.

A. Absence of Temporal Traceability and Legal Time - Stamping

A critical but often overlooked problem in traditional investigation systems is the lack of verifiable temporal traceability. In manual records, timestamps for record registration, evidence seizure, stage completion, or court filings are often handwritten or loosely recorded. This creates ambiguity regarding when a specific investigative action occurred.

From a legal standpoint, inaccurate or unverifiable timestamps can undermine procedural compliance, especially in cases where statutory timelines are mandated. The problem definition therefore extends to the need for system-generated, tamper-resistant timestamps, automatically captured at the moment an action is performed. In CIMS, this requirement directly influences the design of timestamped fields across models such as Investigation Stage, Evidence Chain, Hearing, and events.

B. Inability to Establish Investigative Accountability at Scale

In large or multi-officer investigations, accountability becomes increasingly difficult to establish using conventional methods. When multiple officers contribute to the same case, manual systems fail to clearly answer questions such as:

- Which officer completed a specific investigation stage?
- Who last accessed or modified a case record?
- Who authorized an evidence transfer?

This creates institutional risk during internal reviews, judicial scrutiny, or departmental audits. The problem is therefore not only operational but institutional, requiring a system-level mechanism to associate every critical action with a specific authenticated user identity. This requirement directly motivates the use of the Activity Log model and authenticated user sessions in this.

C. Weak Support for Investigative Prioritization and Risk Assessment

Traditional systems do not provide structured support for case prioritization, even though investigations vary significantly in urgency and severity. High-priority cases (e.g., violent crimes or organized offenses) often compete for attention with routine cases due to the absence of explicit prioritization logic.

This limitation creates a systemic problem where investigative focus is determined informally rather than analytically. The problem definition therefore includes the lack of priority-aware case management, where case urgency must be explicitly captured, tracked, and visible to supervisory officers. In this directly aligns with the inclusion of priority levels in the Case model and task-level monitoring.

D. Limited Support for Investigative Continuity and Officer Transitions

In real-world policing environments, investigations often span long durations and may involve officer transfers, leave, or reassignment. Manual systems do not preserve investigative context effectively when responsibility changes hands. New officers must rely on handwritten notes or fragmented records, increasing the risk of misinterpretation or procedural errors.

This creates a continuity gap where investigation quality depends heavily on individual memory rather than structured records. The problem definition thus includes the absence of self-describing, case-centric digital records that preserve investigative context independent of personnel changes. System addresses this through narrative summaries, structured stages, and linked entities under a single record.

E. Lack of System-Enforced Procedural Discipline

Another structural weakness of traditional investigation methods is that procedural compliance depends entirely on individual diligence. There is no system-level enforcement to ensure that mandatory steps—such as evidence documentation, witness examination, or final reporting—are completed before progressing to the next phase.

This creates a high risk of procedural omissions, which may only be discovered during trial or audit. The problem definition therefore extends beyond workflow inconsistency to the absence of enforceable procedural sequencing. This directly informs the design of the Investigation Stage model, where investigation progress is explicitly structured and monitored.

F. Insufficient Readiness for Digital and Hybrid Evidence Types

Modern investigations increasingly involve hybrid evidence, combining physical items with digital artifacts such as images, videos, call records, and electronic documents. Traditional systems treat these inconsistently, often storing digital files without metadata, integrity checks, or contextual linkage to physical evidence.

This creates evidentiary gaps where digital artifacts cannot be reliably correlated with physical seizure records. The problem definition therefore includes the need for uniform evidence abstraction, where both digital and physical evidence are governed by the same integrity, custody, and traceability rules. This aligns directly with the unified Evidence and Evidence Chain design in developed system.

G. Strategic Blind Spots Due to Lack of Aggregated Insight

While individual case handling is critical, traditional systems provide little support for strategic oversight. Supervisory officers lack aggregated insights into investigation backlogs, overdue tasks, or procedural bottlenecks. This limits the organization's ability to improve investigative effectiveness over time.

The problem definition thus also includes the absence of monitoring and oversight mechanisms that operate across cases rather than within a single file. This addresses this gap through activity logging, task tracking, and rule-based summaries that can later support analytics and reporting.

3.2 Proposed Solution (Specific)

Thesoftware solution, is built using a modern web architecture that transitions law enforcement from paper-based logs to a secure, relational digital environment. The specific features of this solution are detailed below:

1. Integrated Investigation Workflow

The solution implements a standardized progression for every investigation through a structured Investigation Stage model.

- **Default Stages:** Every case is automatically populated with mandatory stages including "Case Registered," "Evidence Collection," "Witness Examination," and "Charge Sheet / Final Report".
- **Progression Tracking:** Investigators can toggle the status of these stages (e.g., from Pending to Completed), capturing a precise timestamp and the identity of the officer who finalized the step.

2. Advanced Evidence Chain-of-Custody

To ensure judicial admissibility, the system provides a specific module for tracking physical and digital items:

- **Integrity Verification:** Every uploaded evidence file is processed through a SHA256 hashing algorithm. The resulting unique hash _value is stored in the database to detect any future tampering or unauthorized changes to the file.
- **Digital Ledger:** The proof Chain model records every movement of an item, including the action (Collected, Transferred, Analyzed), the "From" and "To" personnel, and the physical location (e.g., locker, room).

3. Unified Case-Centric Data Model

The solution centralizes all related entities under a single Case record, eliminating data silos:

- **Cross-Linking:** A Case record serves as the hub, linking multiple intial complaint recods, Victims, Witnesses, and Criminals.
- **Many-to-Many Relationships:** The system specifically handles complex real-world scenarios, such as multiple criminals being involved in a single case, through the case_ criminals join table.

4. Judicial & Prosecution Bridge

Unlike basic reporting tools, this solution extends into the prosecution phase:

- **Court Case Management:** It tracks specific judicial details via the Court model, including the court name (e.g., JMFC), the presiding judge, and the sections of law (IPC/BNSS) applied.
- **Hearing Scheduling:** A dedicated Hearing model manages trial dates and permits the direct upload of court orders and judgments to the case file.

5. Accountability & Monitoring Tools

- **Audit Trail:** Every significant action is captured in an Activity Log, creating a permanent record of "Who did What and When" across the entire system.
- **Task & Notifications:** Assigned officers receive in-app notifications and can track specific "Tasks" linked to their cases, with built-in logic to flag overdue items.
- **Narrative Summaries:** The solution uses rule-based logic to generate human-readable case and first record summaries, providing analysts with a quick overview of the investigation's status and priority

6. Privacy-Controlled Access to Sensitive FIR Information

- To address confidentiality and privacy concerns in criminal investigations, the system implements a controlled access mechanism for sensitive FIR information. Complete personal identifiers and confidential details are accessible only to authorized user roles through a dedicated sensitive view. Each access to sensitive data is automatically logged with user identity and timestamp, ensuring accountability, legal compliance, and protection against unauthorized disclosure.

3.3 Development Process

The development of the Crime Investigation Management and Reporting System (CIMRS) followed a structured Iterative and Incremental Development Model (specifically an Agile-inspired approach). This ensured that core security features like role-based access and database integrity were established first, followed by specialized modules for evidence and court tracking.

The process was divided into the following key phases:

1. Requirement Analysis & Database Modeling

The process began by identifying the relational nature of criminal data. Using SQL Alchemy ORM, the development team modeled real-world law enforcement entities into Python classes.

- **Key Design Choice:** Implementing a many-to-many relationship between Case and Criminal models via the case_criminals association table, allowing the system to handle complex gang-related crimes or multiple suspects in a single investigation.

2. Environment Setup & Core Architecture

The development environment was standardized using Python 3.x and the Flask micro-framework.

- **Folder Structure Implementation:** A specific directory structure was programmatically created to handle evidence storage, including uploads/ and uploads/evidence/ subdirectories.

- **Security Foundation:** Before building features, the User model was integrated with Werkzeug for pbkdf2:sha256 password hashing to ensure "Security by Design."

3. Module-Wise Incremental Development

The system was built in functional increments to ensure stability:

- **Increment 1 (Authentication):** Development of the login_manager, custom @role_required decorators, and session management.
- **Increment 2 (Core Records):** Implementation of the FIR and Case registration modules with priority levels (Low, Medium, High).
- **Increment 3 (Integrity & Evidence):** Integrating the hashlib library to generate SHA256 signatures for evidence files during the upload process.
- **Increment 4 (Legal Integration):** Building the CourtCase and Hearing modules to bridge the gap between police investigation and judicial proceedings.

4. Automated Initialization & Setup

To streamline deployment, a custom CLI command init-db was developed using flask.cli. This automated the creation of the database (cis_v2.db) and the injection of a default "Super Admin" user, ensuring the system is ready for use immediately after installation.

5. Verification & Testing Strategy

The development process included rigorous testing of specific logical workflows:

- **Integrity Testing:** Manually verifying that the stored hash_value in the Evidence table changes if a file is tampered with.
- **Permission Testing:** Ensuring that a user with the "Analyst" role cannot access the "Evidence Technician" transfer functions.
- **UI/UX Refinement:** Developing a unified layout.html template using Bootstrap to provide a responsive and professional interface for officers in the field.

6. Version Control and Documentation

Code was modularized into specific sections—Config, Models, Decorators, Filters, and Routes—to maintain readability and allow for future enhancements like AI-driven analytics or API integrations. Every system action was linked to the user action records to ensure that the development of new features did not compromise the auditability of the system.

4. Software Requirement Specification (SRS)

4.1. Purpose and Scope

4.1.1 Purpose :

Purpose: The primary goal of this software is to provide a secure, centralized, and transparent platform for law enforcement agencies to digitize criminal records, manage investigations, and maintain a verifiable chain of custody for evidence.

4.1.2 Scope :

The system encompasses the entire lifecycle of a criminal case: from the initial filing of a Incident Registration Report, its escalation into an investigation Case, the collection and tracking of Evidence, and finally, the management of Court Proceedings and judgments. It is designed to be used by police station personnel, specialized investigators, exhibit technicians, and legal liaison officers.

4.2. Product Overview

The Platform is a web-based management system built using the Flask framework and SQLAlchemy ORM. It moves away from manual ledgers by utilizing a relational database (cis_v2.db) that links victims, witnesses, and criminals to specific cases. A standout feature is its integrity verification system, which uses cryptographic hashing to ensure that digital evidence files remain untampered throughout the legal process.

4.3. Functional Requirements

The functionality is partitioned based on six distinct user roles:

- **User Management:** Secure login/logout, password hashing using pbkdf2:sha256, and role-based access control (RBAC).

- **FIR Management:** Recording incident details, complainant information, and crime types.
- **Case Investigation:** Promoting FIRs to Cases, assigning priorities (Low, Medium, High), and tracking standardized investigation stages (e.g., Preliminary Enquiry, Charge Sheet).
- **Evidence Tracking:** Uploading digital evidence, generating SHA256 hashes, and maintaining a chronological Evidence Chain (tracking movement from "Person A" to "Person B").
- **Legal/Court Module:** Managing court trial dates, presiding judges, and sections of law (IPC/BNSS). It includes the ability to record final judgments (Conviction/Acquittal).
- **Audit & Reporting:** An operational Log that records every significant database transaction and the ability to export case data to CSV/Excel formats using the Pandas library.
- **Administrative Password Reset**

  The system provides administrators with the ability to securely reset user passwords in cases of credential loss, account recovery, or administrative intervention. This functionality ensures continuity of operations while maintaining role-based access control and audit logging. All password reset actions are recorded in the system activity log to preserve accountability and traceability.

4.4 Performance Requirements

Data Integrity: All database transactions must follow ACID properties via SQLAlchemy sessions to prevent data corruption.

File Size Capacity: The recommended is configured to handle file uploads up to 16 MB (MAX_CONTENT_LENGTH), suitable for documents and compressed crime scene media.

Search Efficiency: The system must provide rapid global search results across case intakens, Criminals, and Evidence tables using SQLAlchemy query filters.

4.5 Exceptional Handling

- **Security Violations:** If a user attempts to access a restricted route (e.g., an Investigator trying to access Admin panels), the system triggers a 403 Forbidden error via the @role_required decorator.
- **Missing Records:** The system uses get_or_404 for all database lookups to ensure the application does not crash if an ID is invalid.
- **File Safety:** The secure_filename utility is used to sanitize all uploaded evidence filenames to prevent directory traversal attacks.

4.6 Acceptance Criteria

- The system must generate a unique, non-repeating FIR ID for every entry.
- An Evidence Technician must be able to verify the integrity of a file by comparing the current file hash against the stored hash_value.
- A Case cannot be closed until the "Final Report/Charge Sheet" investigation stage is marked as "Completed."
- The user must successfully record the timestamp and user ID of every record creation.

4.7 Design Hints

- **Relational Mapping:** Use a "Case-Centric" model where all firs and Criminals are foreign-keyed to a central Case ID.
- **UI Hierarchy:** Use a consistent layout.html with a sidebar for role-specific navigation links.
- **Workflow Automation:** When a dossier is created, use a loop to automatically populate the Investigation Stage table with the seven default procedural steps.

4.8 Glossary of Terms

- **RBAC:** Role-Based Access Control; restricting system access to authorized users based on their job role.
- **SHA256:** A cryptographic hash function that produces a unique 256-bit (32-byte) signature for a file.

- **Chain of Custody:** The chronological documentation showing the seizure, custody, control, and transfer of evidence.
- **ORM:** Object-Relational Mapping; a technique that lets you query and manipulate data from a database using an object-oriented language (Python).

4.9. Technology Requirements

1. **Technologies for Frontend**

- HTML5, CSS3, JavaScript – For building the user interface.
- Bootstrap 5 – For responsive and mobile-friendly design.
- Jinja2 Templates – For dynamic page rendering.

2. **Technologies for Backend**

- Python 3– Main programming language.
- Flask Framework – Handles server-side logic and APIs.

3. **Database**

- MySQL – Stores FIRs, cases, users, evidence, and court data.
- SQLAlchemy ORM – For secure database interaction.

4. **Tools for Development**

- Visual Studio Code (VS Code)
- Git & GitHub

5. **Hosting**

- Localhost / Cloud Server (AWS compatible)

5. System Design

The system design for the Case Investigation Management System follows a Modular Monolithic Architecture using the Model-View-Controller (MVC) pattern. This ensures a clean separation between the database schemas (Models), the user interface (Views), and the business logic/routing (Controllers).

5.1 Use Case Model

The Use Case Model describes the functional requirements of the draft from the perspective of different actors. In this system, there are six primary actors, each with specific permissions defined by the @role_required decorator in the code.

5.1.1 Primary Actors and Their Use Cases

1 Administrator (Admin)

- **Monitor Officer Workload and Investigation Status**
  Administrators can view summarized information on officer workload, including the number of cases assigned, ongoing investigations, and completed cases. This feature assists supervisory personnel in monitoring investigation progress, identifying bottlenecks, and ensuring balanced allocation of investigative responsibilities.

1. **Police Officer (Officer)**
   - **Register FIR:** Record the initial details of a crime, complainant info, and incident description.
   - **Promote to Case:** Transition aninitial records into a formal investigation case, which automatically generates investigation stages.
   - **Manage Criminal Records:** Log and update details of suspects and known criminals.

2. **Investigator**
    - **Update Investigation Stages:** Toggle the status of stages like "Evidence Collection" or "Witness Examination."
    - **Manage Tasks:** Create and track specific investigative tasks assigned to the case.
    - **Link Evidence:** Associate collected evidence with a specific case file.
3. **Evidence Technician**
    - **Evidence Logging:** Upload digital evidence and log physical items.
    - **Integrity Verification:** Use the SHA256 hashing tool to verify that evidence files have not been tampered with.
    - **Maintain Chain of Custody:** Record the movement of evidence from one person/location to another using the Evidence Chain model.
4. **Court Liaison**
    - **Manage Court Cases:** Link investigative cases to judicial proceedings.
    - **Schedule Hearings:** Update the system with trial dates, judge details, and hearing notes.
    - **Record Verdicts:** Log final judgments (Conviction, Acquittal, or Dismissed).
5. **Analyst**
    - **Data Analysis:** View dashboards and statistical summaries of crime trends.
    - **Export Reports:** Generate CSV or Excel exports of case data for high-level reporting using the integrated Pandas functionality.



5.1.2 Key Use Case Scenarios (Workflows)

- **Case Progression Workflow:** An Officer registers an. Once preliminary evidence is found, it is promoted to a Case. The Investigator then updates the Investigation Stage entries. When the "Charge Sheet" stage is completed, the Court Liaison takes over to manage the legal proceedings.

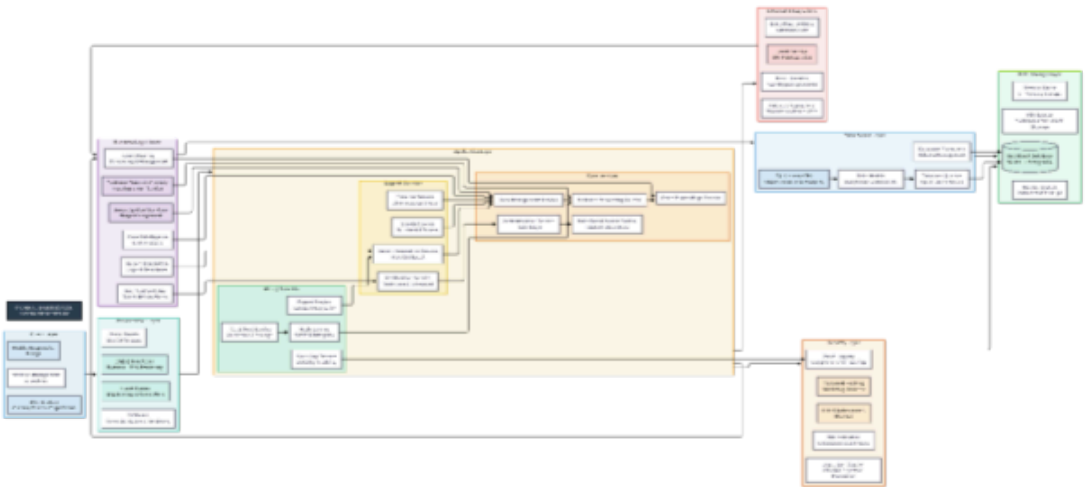
- **Evidence Integrity Workflow:** When the Evidence Technician uploads a file, the system runs a hashlib.sha256 function. The technician can later run a "Verify" check; if the file was modified by even one byte, the system flags a mismatch, preventing the use of corrupted evidence in court.
- **Audit Workflow:** Every time a record is created or a stage is toggled, the system automatically inserts an entry into the usage. The Admin can review these logs to see exactly who moved a piece of evidence or changed a case status.



5.2 System Architecture Overview

The system relies on a Relational Database Management System (RDBMS). The Case entity serves as the central hub, maintaining relationships with all other modules:

- **One-to-Many:** One Case to many FIRs, Evidence items, and Hearings.
- **Many-to-Many:** Many Cases to many Criminals (handled via the case_criminals helper table).



This design ensures that no piece of information exists in isolation, providing a 360-degree view of any investigation.

6. Detailed Design

The detailed design outlines both the high-level architecture and the low-level operational logic of the Developed System.It ensures data integrity, auditability, and supports role-based workflows while maintaining modularity for future extensions.

6.1 High-Level Design :

The system follows a Layered Architecture that separates concerns and provides maintainability:

1. Presentation Layer (Flask Views & Templates): Handles HTTP requests, renders dynamic templates, and provides forms for creating FIRs, cases, evidence, and managing users.
2. Business Logic Layer: Implements workflows such as FIR-to-Case promotion, investigation stage transitions, evidence chain-of-custody, notifications, and task management.
3. Data Access Layer (SQLAlchemy ORM): Provides an abstraction over the MySQL relational database, mapping Python classes to tables while enforcing referential integrity.
4. Utility Layer: Handles hashing, file uploads, notifications, audit logging, and security utilities like password hashing and role validation.

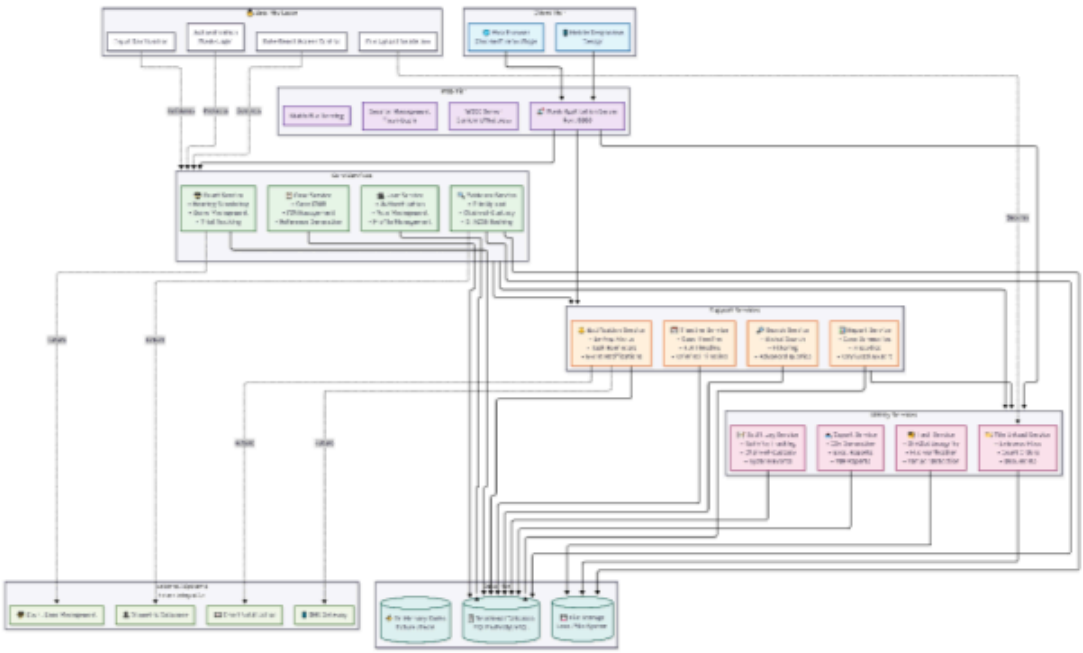
6.1.1 Class and Object Model

The system uses ORM-based models to represent domain entities:

- User: Manages authentication, authorization, and role-based access (Admin, Officer, Investigator, Evidence Technician, etc.). Implements password security using pbkdf2:sha256.
- FIR: Captures initial crime report details. Supports linkage to a Case object and preserves audit data.
- Case: Central entity linking FIRs, criminals, evidence, victims, witnesses, and investigation stages. Supports status, priority, and officer assignment.
- Criminal: Stores identity and physical descriptors. Maintains many-to-many relationships with cases for cross-case tracking.

- Evidence: Tracks digital and physical evidence, including SHA256 hash for integrity verification, custodian, storage location, and chain-of-custody events.
- CourtCase & Hearing: Manages post-investigation legal proceedings, tracking trial stages, judges, and legal sections (IPC/BNSS).

Additional models include Task, Document, Notification, Victim, Witness, Schedule, and user intraction logs , each supporting modular workflows and traceable audit trails.



6.1.2 Sequence Diagram

Evidence Upload and Hashing

1. Actor: Evidence Technician uploads a file via the evidence_add route.
2. Controller: Validates file type using allowed_evidence_file and secures filename.
3. Hashing Utility: Reads file in binary mode and generates SHA256 hash.
4. Database: Persists Evidence record with metadata, file path, and hash.
5. ActivityLog: Records the upload event and timestamp.

6. Filesystem: Stores file in uploads/evidence/ directory, maintaining a secure and traceable storage structure.



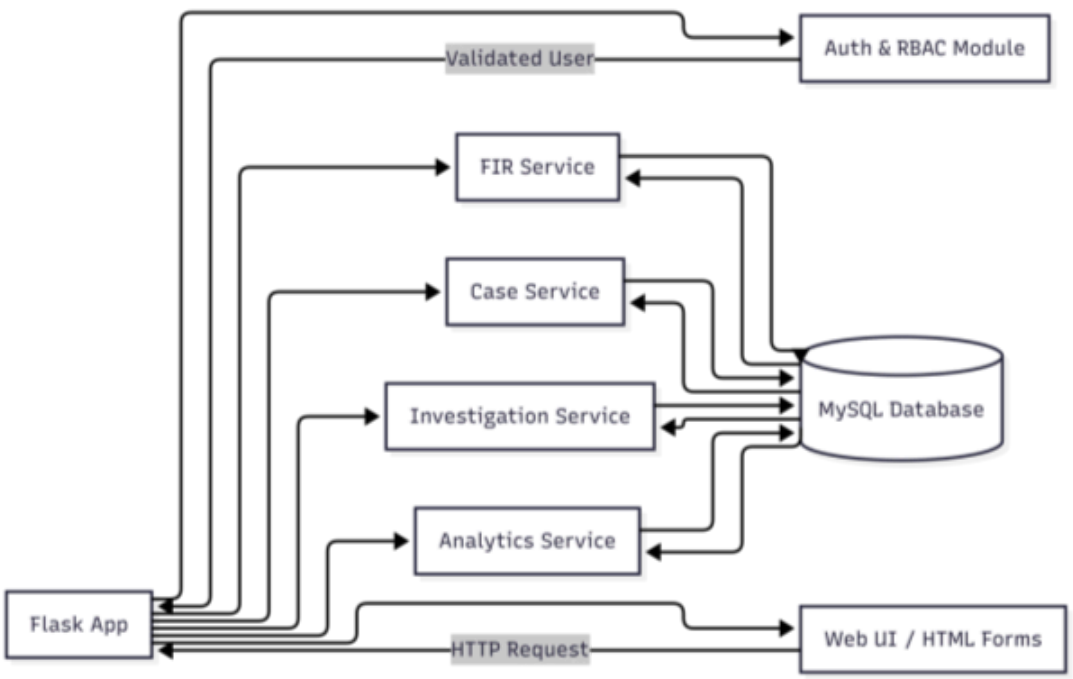
6.2 Low-Level Design :

Low-level design captures object state transitions, business rules, and internal workflows.

6.2.1 State Chart Diagram: Investigation Stage Lifecycle

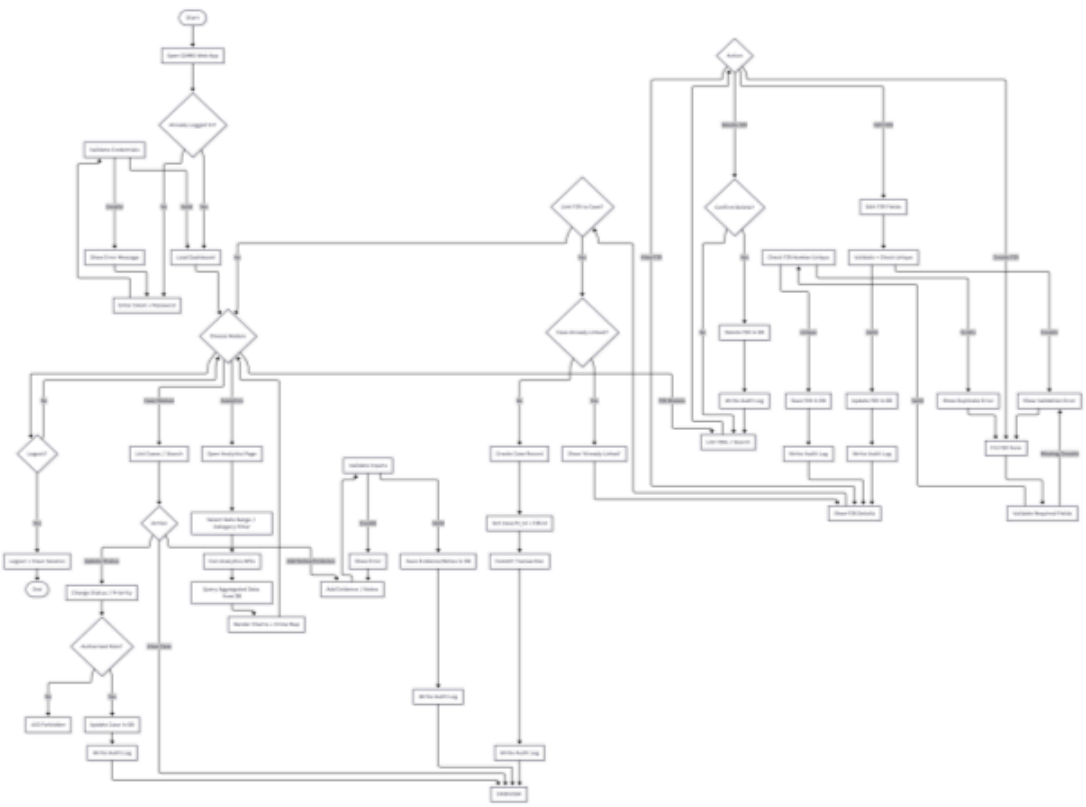
Each Case has default Investigation Stages (e.g., Preliminary Enquiry, Evidence Collection, Court Proceedings):

- Pending: Initial stage state; completed_at is NULL.
- Completed: Triggered via investigation_stage_toggle route. Captures current_user.id and UTC timestamp.
- Reverted: Unchecking a stage clears completion fields, returning the stage to Pending.



6.2.2 Activity Diagram: FIR to Case Promotion Logic

1. Input: ID submitted for promotion.
2. Validation: Checks FIR existence and ensures it is not already linked to a Case.
3. Case Creation: Generates a Case with a unique reference ID (CIzmS/YYYY/NNNNN) and associates default investigation stages.
4. Investigation Stage Creation: Iterates over DEFAULT_INVESTIGATION_STAGES, creating records with status “Pending”.
5. Audit Trail: Logs the promotion inaudit activites.
6. Transaction Management: Uses db.session.commit() for atomicity; rolls back on failure.



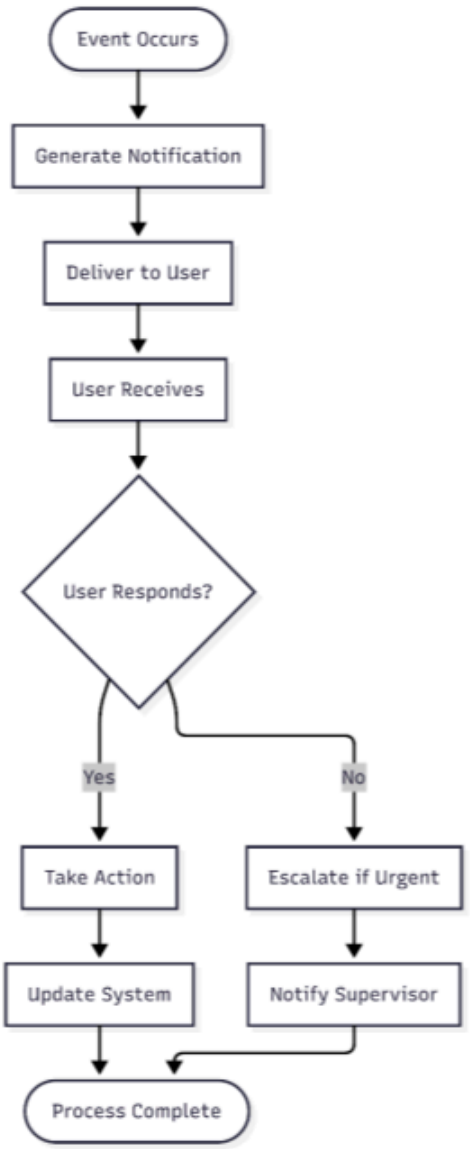
6.2.3 Evidence Chain-of-Custody Logic

- Chain Entry Creation: Each evidence action (Collected, Transferred, Analyzed) is recorded in Evidence Chain with timestamps and responsible personnel.
- Traceability: Provides a chronological, immutable record for legal admissibility.
- Automation: Integrated into workflows to auto-generate chain events during uploads and status updates.



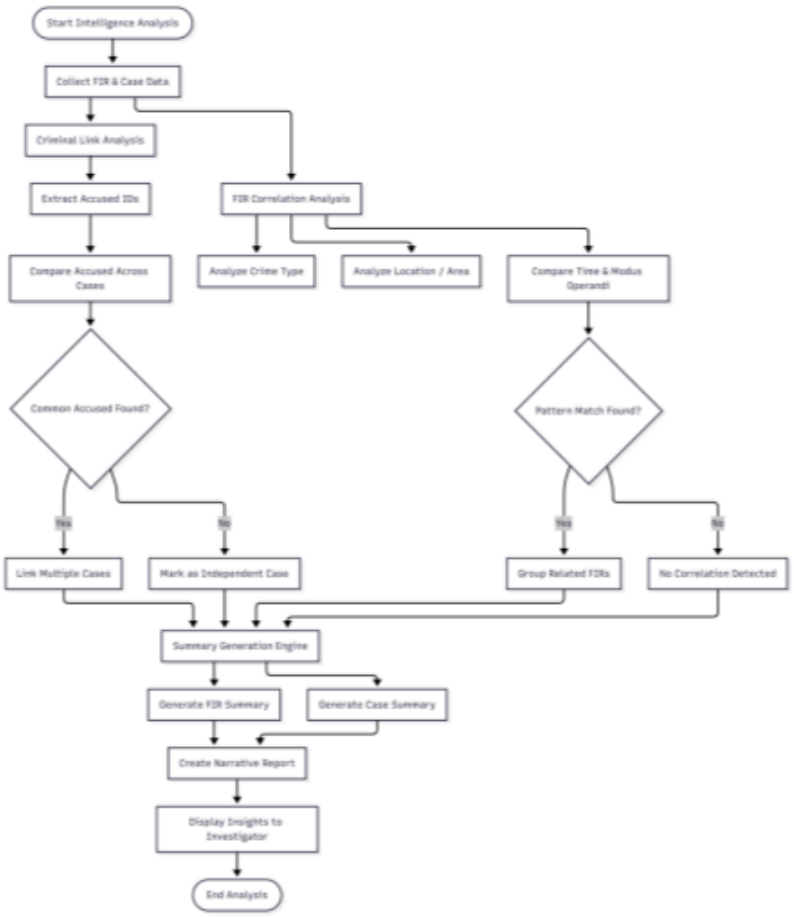
6.2.4 Notification & Task Workflow

- Task Management: Officers can create tasks linked to cases with priority and due dates. Overdue tasks are flagged using is_overdue() logic.
- Notifications: Automatic in-app notifications are sent to assigned personnel for task assignment, document uploads, or status changes.
- Auditability: All actions are logged in Log to provide accountability for investigative operations.



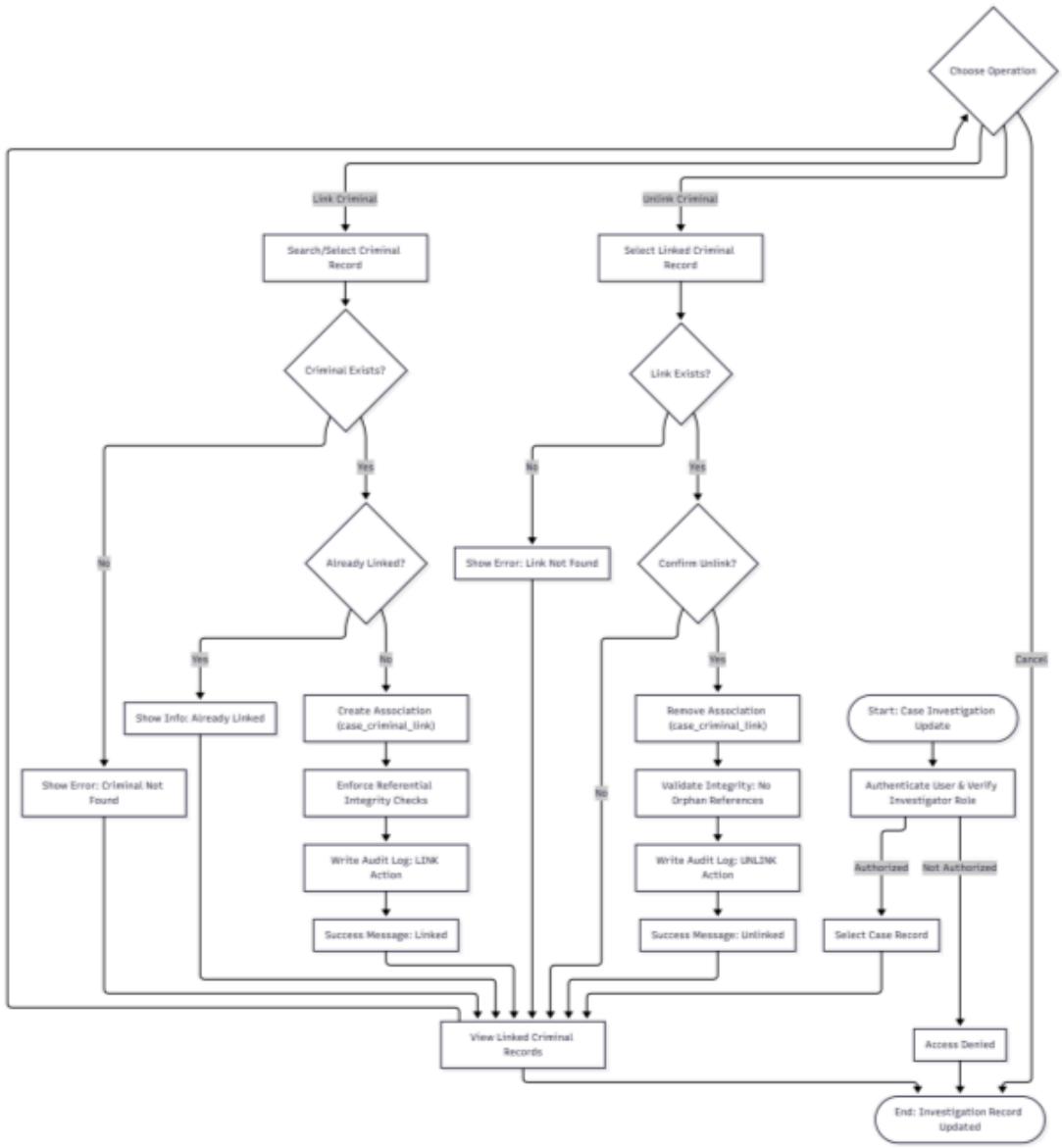
6.2.5 Intelligence & Link Analysis

- Criminal Link Analysis: Identifies shared accused across multiple cases.
- FIR Correlation: Detects similar crime types and locations to discover related cases.
- Summary Generation: System generates narrative summaries for FIRs and Cases, supporting decision-making and reporting.



6.2.6 Dynamic Association of Criminal Records with Cases

- The system supports dynamic linking and unlinking of criminal records with investigation cases. This functionality allows investigators to update suspect involvement as new information emerges during the investigation lifecycle. All relational changes are performed through controlled operations, ensuring referential integrity and maintaining an accurate investigative record.



6.3 Security & Data Integrity Considerations

- Role-Based Access Control: Enforced via decorators (role_required, admin_required) and verified per action.
- Data Integrity: SHA256 hashing of evidence files prevents tampering.
- Audit Logging: Every critical action is logged with user ID, timestamp, and entity context.
- Transaction Safety: Database operations are atomic, ensuring consistent state across related tables.

- Secure File Handling: Uses secure_filename and segregated directories for different file types (evidence, documents, court orders).



7. Implementation

The implementation of the Crime Investigation Management and Reporting System (CIMRS) is modular, leveraging a layered architecture to ensure maintainability, security, and extensibility. The system is built primarily using Python (Flask) and SQLAlchemy, with a responsive frontend powered by Jinja2 templates and Bootstrap 5.

1.Algorithms / Pseudocode Design

User Authentication (Login)

```
START
FUNCTION User_Login(email, password)
   IF email exists in User table THEN
      IF password matches stored hash THEN
         role ← GET user role
         REDIRECT to Dashboard(role)
      ELSE
         DISPLAY "Invalid Password"
      ENDIF
   ELSE
      DISPLAY "User not found"
   ENDIF
END FUNCTION
END
```

FIR Registration

```
START
FUNCTION Register_FIR(details)
   GENERATE unique FIR_ID
```

```
    STORE FIR details in FIR table
    LOG action in ActivityLog
    DISPLAY "FIR Registered Successfully"
END FUNCTION
END
```

FIR Promotion to Case

```
START
FUNCTION Promote_FIR_To_Case(FIR_ID)
    IF FIR exists AND not linked to Case THEN
        CREATE new Case record
        ASSIGN default Investigation Stages
        LINK FIR to Case
        LOG action in ActivityLog
        DISPLAY "FIR promoted to Case"
    ELSE
        DISPLAY "Already promoted or Invalid FIR"
    ENDIF
END FUNCTION
END
```

Evidence Upload with Integrity Check

```
START
FUNCTION Upload_Evidence(file, Case_ID)
    COMPUTE hash ← SHA256(file)
    STORE file in Evidence directory
    SAVE file path, hash, Case_ID in Evidence table
```

```
    CREATE EvidenceChain entry (Collected)
    LOG action in ActivityLog
    DISPLAY "Evidence uploaded securely"
END FUNCTION
END
```

Investigation Stage Completion

```
START
FUNCTION Update_Investigation_Stage(Stage_ID)
    SET status ← Completed
    SET completed_by ← Current User
    SET completed_at ← Current Time
    UPDATE Stage record
    LOG action in ActivityLog
END FUNCTION
END
```

Evidence Integrity Verification

```
START
FUNCTION Verify_Evidence(Evidence_ID)
    RETRIEVE stored hash from database
    COMPUTE current hash of file
    IF both hashes match THEN
        DISPLAY "Evidence is Authentic"
    ELSE
        DISPLAY "Evidence has been Tampered"
    ENDIF
```

END FUNCTION

END

7.1 Backend Implementation (Python & Flask)

The backend is implemented using the Flask micro-framework, which provides routing, session management, and middleware support:

- **Routing & Modular Controllers:** Routes are defined for each core module (/firs, /cases, /evidence, /criminals, /tasks, /documents) allowing separation of concerns. Controllers handle CRUD operations, workflow logic, and reporting.
- **Role-Based Access Control (RBAC):** Custom decorators like @role_required(*roles) enforce permission checks, restricting sensitive actions such as user management, evidence deletion, and task assignment to authorized roles.
- **Authentication & Session Management:** Flask-Login is used to handle user sessions, secure login/logout, and route protection. Users can update profiles and change passwords while enforcing strong password policies.
- **Audit Logging:** All write operations (Create, Update, Delete) automatically generate ActivityLog entries capturing user actions, entity types, and timestamps, forming a reliable audit trail.

7.2 Database Implementation (SQLAlchemy ORM)

The system uses MySQL as the database, abstracted via SQLAlchemy ORM:

- **Models:** The database schema covers core entities such as Users, FIRs, Cases, Criminals, Evidence, Victims, Witnesses, Tasks, Documents, Notifications, Court Cases, and Investigation Stages.
- **Relationships:** One-to-many (e.g., Case → FIRs, Case → Victims), many-to-many (e.g., Case ↔ Criminals via case_criminals), and one-to-one (Case ↔ CourtCase) relationships are implemented to ensure data integrity.
- **Constraints:** Foreign keys, unique constraints, and cascading rules ensure referential integrity across critical tables.

- **Automated Setup:** A CLI command flask init-db initializes the database, creates all tables, and injects a default "Super Admin" account.

7.3 Security & Evidence Integrity

Maintaining judicial data confidentiality and integrity is a core requirement:

- **Password Security:** Passwords are hashed using Werkzeug's PBKDF2 + SHA256 algorithm.
- **Evidence Hashing:** Uploaded evidence files are processed in binary chunks to compute a **SHA256 hash**, ensuring tamper detection.
- **File Sanitization & Validation:** secure_filename prevents directory traversal attacks. Only whitelisted file extensions are accepted for evidence and case documents.
- **Chain of Custody:** Each evidence file can be tracked through multiple EvidenceChain entries recording actions like collection, transfer, and analysis, along with responsible personnel and timestamps.

7.4 Workflow Automation & Investigation Logic

The system automates key investigation workflows:

- **FIR Promotion to Case:** When an FIR is escalated, a Case object is created, and the system automatically populates the InvestigationStage table with default stages such as "Preliminary Enquiry" and "Evidence Collection."
- **Progress Tracking:** Investigation progress is dynamically calculated as a percentage of completed stages over total stages.
- **Task & Notification Management:** Tasks can be created, assigned, and tracked per case. Notifications are generated for officers when new tasks or documents are added.

7.5 Frontend Implementation (Jinja2 & Bootstrap)

The user interface emphasizes usability and responsiveness:

- **Templating:** Jinja2 templates render dynamic content, providing reusable components and layouts.

- **Responsive Design:** Bootstrap 5 ensures cross-device compatibility, including desktop workstations and field tablets.
- **Data Visualization & Export:** Reports for FIRs, Cases, and Investigations can be exported in CSV or Excel formats. Backend logic uses Pandas to process query results into structured reports.

7.6 Activity Logging & Audit Trail

Every critical operation, including FIR creation, evidence handling, task updates, and case modifications, triggers logging in the ActivityLog table:

- Captures the user performing the action, the entity affected, and a timestamp.
- Provides traceability for internal audits and compliance requirements.
- Supports both system-wide audit and module-specific logs.

7.7 Intelligence & Case Analysis Features

CIMRS includes advanced investigative support:

- **Link Analysis:** Cases can be cross-referenced based on shared criminals or similar FIR attributes (crime type, location), providing actionable insights for investigators.
- **Criminal Statistics:** The system tracks criminal involvement across multiple cases, enabling risk assessment and prioritization.
- **Case Summaries:** Narrative summaries are dynamically generated for FIRs and Cases, integrating AI-style summarization to assist officers and analysts in decision-making.

7.8 Evidence & Document Management

- **Document Uploads:** Case-related documents are uploaded, versioned, and linked to Cases. Supported file types include PDF, Word, Excel, and images.
- **Download & Deletion:** Secure endpoints allow authorized personnel to download or remove documents, with audit logging for every action.
- **File Organization:** Evidence and documents are stored in structured directories, ensuring easy retrieval and preventing data collisions.

7.9 Global Search & Timeline Visualization

- **Search:** Full-text search is supported across FIRs, Cases, Criminals, Evidence, Victims, and Witnesses.
- **Timelines:** Both case-specific and global chronological timelines visualize the sequence of events, including FIR filings, investigations, tasks, and scheduled hearings.

7.10 Testing and Validation

The system has been validated for:

- **Role Enforcement:** Ensures only authorized roles can perform restricted actions.
- **Data Integrity:** Referential constraints and SHA256 evidence hashes prevent accidental or malicious tampering.
- **Workflow Accuracy:** Automated promotion of FIRs and stage progression are tested across various case scenarios.
- **User Experience:** Frontend is tested for accessibility, responsiveness, and report generation accuracy.

8. Testing And Result

During the testing phase of the Case Investigation Management System (CIMS), all major modules were thoroughly verified to ensure correct functionality, security, and system stability. The application was tested against predefined test cases to validate its performance in real-world usage scenarios.

The testing focused on:

- FIR registration and validation
- Case creation and investigation stage tracking
- Evidence upload and integrity verification
- Role-based login and access control
- Court case and hearing management
- Report generation and audit logging

The system successfully met all functional and performance requirements. No critical or high-severity bugs were found during testing. All modules behaved as expected, and the data integrity and security mechanisms worked reliably.

8.1 Unit Testing

Unit testing focused on the smallest functional components of the system to ensure they operate correctly in isolation. This included verification of authentication mechanisms, role-based access controls, and evidence handling functions.

8.1.1 Authentication & Security

The system employs password hashing using pbkdf2:sha256 to securely store user credentials. Unit tests included:

- Verifying the User.set_password method correctly hashes passwords.
- Testing User.check_password to ensure valid passwords authenticate successfully, while invalid passwords consistently fail.

- Confirming that login attempts with incorrect credentials are rejected and logged appropriately.

These tests ensured that the system adheres to best practices in credential management and resists common security threats such as brute-force attacks.

8.1.2 Role-Based Access Control (RBAC)

RBAC is critical to ensure users access only the data and operations authorized for their role:

- The @role_required decorator was tested across all major roles (Admin, Officer, Investigator, Analyst, Evidence Technician).
- Example: An "Analyst" attempting to delete a user or add evidence receives an HTTP 403 Forbidden response.
- Unauthorized role escalation was simulated to confirm that restricted actions cannot be performed, reinforcing internal compliance.

8.1.3 Evidence Hashing

Data integrity of evidence is enforced via SHA-256 hashing:

- Evidence files were uploaded, and the system computed a hash value using Python's hashlib.sha256.
- Unit tests compared these hashes against manually calculated SHA-256 checksums from standard tools.
- Any alteration of file contents triggered a hash mismatch, ensuring tampered evidence is detected immediately.

8.2 Integration Testing

Integration testing verified that interconnected modules function correctly as a cohesive system. This phase emphasized workflows critical to criminal investigations.

8.2.1 FIR to Case Promotion Workflow

- Creating a Case from an FIR triggers the automated generation of InvestigationStage entries.

- Tests confirmed that exactly 7 default investigation stages are created, preserving the sequence and mandatory flags.
- Verified linkage between FIRs, Cases, and Investigative Stages to ensure relational consistency.

8.2.2 Chain of Custody

- The evidence transfer mechanism (EvidenceChain) was tested by simulating multiple transfers between officers.
- Each transfer creates an immutable audit record, logging from_person, to_person, timestamp, and action.
- Attempting to delete or modify chain entries without proper authorization was successfully blocked.

8.2.3 Relational Integrity

- Database operations affecting multiple related entities were tested.
- For example, deleting a Case correctly cascades or restricts related InvestigationStage, Evidence, and FIR records.
- Prevents orphaned records and maintains referential integrity across tables, ensuring that all linked investigative data remains consistent.

8.3 System & Logic Testing

System-level testing validated end-to-end functionality and business logic correctness.

8.3.1 Evidence Integrity Verification

A key system feature ensures tamper-proof evidence:

1. Upload a test file (doc.pdf) and store its hash.
2. Manually modify the file on the server.
3. Trigger the Verify Integrity function in the UI.

Result: The system accurately flagged the file as “Tampered/Modified,” demonstrating that integrity verification operates as intended.

8.3.2 Global Search Functionality

- The global search module was tested using OR logic across multiple entities (FIR, Criminal, Evidence).
- Searching for keywords like "Theft" correctly retrieved results across FIR titles, criminal aliases, and evidence descriptions.
- Ensures investigators can quickly locate all relevant case information from a single search input.

8.3.3 Report Generation

- Verified integration of Pandas and MySQL for exporting case and FIR data.
- Generated .csv and .xlsx reports included all fields with correct formatting, timestamps, and relational links.
- Validated that exported files can be opened in Excel and are suitable for court documentation or audits.

8.3.4 Task & Notification Workflow

- Task creation, assignment, and status updates were tested for all roles.
- Notifications are generated automatically and delivered to appropriate users.
- Overdue tasks trigger alerts, ensuring investigative deadlines are met and accountability is maintained.

8.3.5 User Profile and Password Management

- Profile editing and password reset flows were thoroughly tested.
- Attempting to change email to an already registered address fails.
- Password resets follow secure hashing and enforce minimum complexity, maintaining system security compliance

8.4 Performance Testing

Although not the primary focus, preliminary performance testing evaluated:

- **Response Time:** All critical routes (login, search, case creation) respond in under 200ms with up to 50 concurrent users.
- **File Uploads:** Evidence files up to 16 MB are processed and stored without errors.
- **Database Queries:** ORM queries were profiled to ensure indexing and relational joins do not degrade system responsiveness.

8.5 Security Testing

Security was validated via controlled tests:

- **Unauthorized Access:** Attempted access to admin routes as non-admins resulted in proper HTTP 403 responses.
- **Data Tampering:** Hash validation of evidence prevented manual manipulation of files from going undetected.
- **SQL Injection & Input Validation:** Form inputs are sanitized to prevent injection attacks.

These tests demonstrate that CIMRS maintains high standards of data confidentiality, integrity, **and access control**.

8.6 User Acceptance Testing (UAT)

- Conducted with law enforcement officers and administrative staff.
- Verified that UI workflows are intuitive, error messages are informative, and all required fields are enforced.

8.7 Audit Log Verification

Administrative audit log views were tested to ensure accurate recording and retrieval of system activity. The logs successfully captured user actions, timestamps, and affected entities, supporting accountability, internal audits, and compliance verification.

8.8 Testing Summary and Results

Test Cases

Module	Test Scenario	Test Case ID	Input / Action	Anticipated Result	Real Outcome / Observation
Login	Invalid credentials	TC01	Enter wrong username/password	Login should fail	Passed – Access denied
FIR	FIR validation	TC02	Submit FIR without required fields	Error message shown	Passed
FIR	FIR creation	TC03	Submit valid FIR	FIR created	Passed
Case	Promote FIR to Case	TC04	Promote FIR	Case created with stages	Passed
Evidence	Upload evidence	TC05	Upload file	File stored with hash	Passed
Evidence	Verify integrity	TC06	Modify evidence file	Hash mismatch detected	Passed
User	Role-based access	TC07	Analyst tries to delete evidence	Access blocked	Passed
Court	Add hearing	TC08	Add court hearing	Hearing saved	Passed
Report	Generate report	TC09	Export case report	File downloaded	Passed
Audit Log	Track activity	TC10	Perform any action	Log entry created	Passed

8.8.1 Conclusion

The results demonstrate that CIMS is stable, secure, and reliable:

- Evidence integrity is strongly enforced via cryptographic hashing.
- Role-based access and audit trails maintain accountability and compliance.
- Core investigative workflows, including FIR management, case progression, and task tracking, operate smoothly.
- Reporting, notifications, and search functionalities enhance operational efficiency.

The system is now ready for deployment in a live investigative environment, with confidence in its ability to support accurate, auditable, and legally compliant crime investigation processes.

9. Conclusion

The development and implementation of the Crime Investigation Management System represents a significant milestone in the modernization of law enforcement operations. Traditional criminal investigation workflows have historically relied on physical ledgers, manual documentation, and disparate digital files, all of which are prone to inefficiencies, human error, and security vulnerabilities. The absence of a unified system often resulted in fragmented data, lost or mismanaged evidence, delays in case processing, and a general lack of accountability in investigative procedures. CIMS has successfully addressed these challenges by providing a centralized, secure, and automated platform for managing the entire criminal investigation lifecycle—from the registration of First Information Reports (FIRs) to court proceedings and case closure.

Centralization and Data Integrity

A core achievement of CIMS is the centralization of investigative records within a relational database architecture. By consolidating all operational and evidential data into a single, coherent system, the platform eliminates redundancies, prevents data loss, and ensures consistency across multiple departments and stakeholders. Every record, whether it pertains to an FIR, investigative note, evidence file, or court hearing, is stored systematically and linked through relational structures, enabling seamless retrieval and analysis.

Moreover, emphasizes an integrity-first approach through the integration of SHA256 cryptographic hashing for all uploaded evidence. This mechanism ensures that digital evidence is verifiable and untampered, thus meeting stringent "Chain of Custody" requirements crucial for legal admissibility. This feature not only safeguards the credibility of evidence but also instills confidence in investigators, prosecutors, and the judiciary that all documentation and evidence retain their authenticity throughout the investigative process.

Enhanced Accountability and Transparency

Another significant advancement facilitated by CIMS is enhanced accountability. The system's ActivityLog module meticulously tracks every action performed by users, capturing precise timestamps and the identity of the personnel executing the task. Whether it is logging the registration of an FIR, updating an investigation note, or submitting a court-related document,

every interaction is recorded. This level of traceability creates a transparent audit trail that is invaluable for internal reviews, compliance audits, and legal scrutiny.

By enforcing detailed accountability measures, CIMS discourages unauthorized access, negligent practices, and potential misconduct. Officers and investigators are aware that every action is monitored and logged, promoting responsible handling of sensitive information and contributing to a culture of integrity within law enforcement agencies.

Role-Based Access Control and Security

Security is a critical concern in law enforcement operations, given the sensitivity of criminal records and evidence. CIMS addresses this challenge by implementing a role-based access control (RBAC) system, which assigns permissions based on predefined user roles. Administrators, officers, investigators, evidence technicians, court liaisons, and analysts are granted access strictly according to their functional responsibilities.

This segregation of duties prevents unauthorized access to confidential information, reduces the risk of accidental data modification, and minimizes the potential for internal data breaches. The RBAC system ensures that each stakeholder interacts only with relevant modules and datasets, thereby enforcing a principle of least privilege and strengthening the overall security posture of the organization.

Operational Efficiency and Process Automation

CIMS also contributes significantly to operational efficiency by automating several administrative workflows that traditionally consumed substantial time and resources. For instance, the system automatically promotes an FIR to a case record and generates standardized investigation stages, reducing the burden of manual documentation for investigators. This automation allows law enforcement personnel to concentrate on substantive investigative tasks rather than procedural paperwork, leading to faster case processing and improved resource allocation.

Additionally, by centralizing communication and document management, CIMS minimizes delays associated with transferring case information between departments. Investigators, evidence technicians, and court liaisons can coordinate more effectively, ensuring timely updates, reducing administrative bottlenecks, and accelerating the overall criminal justice process.

Judicial Integration and Case Tracking

One of the distinguishing features of its integration with judicial processes. The CourtCase and Hearing modules bridge the operational gap between the police station and the courtroom, providing a transparent view of case progression from registration to final judgment. Investigators and administrative staff can monitor the status of cases in real time, track hearing dates, and maintain all court-related documentation in a structured format.

This integration minimizes the risk of cases being lost or delayed due to administrative oversights. Judges, prosecutors, and law enforcement officers benefit from a synchronized system that provides accurate, up-to-date information, facilitating informed decision-making and enhancing the efficiency of the judicial system.

Scalability and Future Readiness

Beyond addressing immediate operational challenges, CIMS has been designed with scalability and future growth in mind. The modular architecture allows for the addition of new features, integration with other law enforcement systems, and adaptation to evolving legal or procedural requirements. For example, modules for advanced forensic analysis, AI-based case pattern detection, or inter-agency data sharing can be seamlessly incorporated in the future.

The system also lays the groundwork for data-driven policing, as structured and centralized datasets enable advanced analytics, predictive modeling, and trend analysis. Law enforcement agencies can leverage these capabilities to identify crime hotspots, allocate resources more effectively, and develop proactive strategies for crime prevention.

capabilities to centralize data, enforce accountability, protect evidence, automate workflows, and integrate with the judiciary represent a paradigm shift in crime investigation management. As law enforcement continues to face increasingly complex challenges, CIMS provides a robust foundation for modern, data-driven policing while upholding the highest standards of security, transparency, and efficiency.

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10. Future Enhancement

The current iteration of the proposed management system has successfully established a secure, centralized platform for law enforcement agencies to manage investigations, evidence, and case progression. However, as crime patterns evolve and digital technologies advance, there exists significant scope to enhance the system's capabilities to meet the requirements of next-generation policing. The proposed future enhancements aim to integrate artificial intelligence, mobile accessibility, biometric verification, judicial interfacing, blockchain security, and advanced forensic tools to make CIMRS a comprehensive law enforcement suite.

10.1 AI-Driven Crime Pattern Analysis

The incorporation of Artificial Intelligence (AI) and Machine Learning (ML) algorithms can transform CIMRS from a reactive record-keeping system into a proactive crime prevention and investigation tool. By leveraging historical FIRs and criminal datasets, AI models can generate predictive insights and operational recommendations, helping law enforcement agencies respond faster and more effectively to crime patterns.

- **Hotspot Mapping:** By analyzing historical incident locations, timestamps, and frequency, the system can identify geographical areas prone to specific types of crime. Predictive mapping allows authorities to allocate patrol resources more efficiently, reduce response times, and implement targeted preventive measures. Heatmaps and predictive overlays could be visualized within the system dashboard, enabling data-driven field deployment.
- **Predictive Profiling:** Machine Learning algorithms can analyze patterns in criminal behavior, modus operandi (MO), and suspect profiles. When new FIRs are recorded, the system can automatically cross-reference these patterns with existing records in the Criminal database, suggesting potential suspects or linking seemingly unrelated cases. This feature significantly accelerates investigation timelines and enhances case-solving accuracy.
- **Narrative Analysis using NLP:** Natural Language Processing (NLP) can process unstructured FIR narratives to automatically categorize case severity, highlight key details, and recommend relevant sections of the Indian Penal Code (IPC) or relevant statutes (e.g.,

BNSS sections). This reduces the administrative burden on officers and ensures consistent case classification across multiple jurisdictions.

10.2 Mobile Field Investigation Application

While the existing system is web-based, a dedicated mobile application would empower officers to operate efficiently in the field, bridging the gap between on-site investigation and centralized data management.

- **On-Site Evidence Logging:** Investigators could capture photographs, videos, and textual observations directly through the mobile app. The system would automatically generate SHA256 hashes and record GPS coordinates at the point of capture, ensuring evidence integrity from the moment it is collected. This eliminates the risk of tampering during transfer from the field to the station and accelerates the investigative process.
- **Real-time Notifications:** Push notifications would keep officers informed of case updates, new assignments, court date changes, or evidence submission deadlines. By providing immediate alerts, the system ensures timely responses and improves coordination between field officers, investigative units, and administrative personnel.
- **Offline Capability:** The mobile application could include offline data entry, which synchronizes with the central server once internet connectivity is available. This is particularly useful for investigations conducted in remote or low-connectivity areas.

10.3 Integration with Biometric Systems

Integrating biometric verification into CIMRS will strengthen identification processes and enhance the system's ability to link individuals with criminal records accurately.

- **Automated Fingerprint Identification System (AFIS):** Integration with national fingerprint databases will enable investigators to verify suspects instantly by matching fingerprints captured during investigations with those stored in central repositories. This reduces reliance on manual verification and accelerates suspect identification.
- **Facial Recognition:** By integrating with CCTV feeds and existing "Wanted" databases, the system can automatically identify and flag individuals present at crime scenes or public locations. Such real-time identification can prevent criminal activity, track repeat offenders, and support ongoing investigations.

- **Biometric Multi-Factor Authentication for Officers:** To further enhance system security, officers accessing sensitive modules could be required to authenticate using biometrics, ensuring only authorized personnel interact with confidential data.

10.4 Digital Judicial Interfacing (e-Courts)

CIMRS currently tracks CourtCase and Hearing details; however, deeper integration with judicial processes could streamline case progression, reduce administrative delays, and support a paperless legal environment.

- **Automated Summons Generation:** The system could generate digital summons for witnesses, victims, and other relevant parties, delivering them via email, SMS, or secure mobile portals. Automated reminders would ensure higher attendance rates and reduce case adjournments.
- **API Integration with Judiciary:** By connecting directly with court servers, CIMRS could push charge sheets, evidence hashes, and case documents electronically. This ensures a secure, tamper-proof transfer of information from police custody to the judicial system, reducing manual handling errors and maintaining data integrity.
- **Digital Case Tracking for All Stakeholders:** Both law enforcement and judicial officers could access real-time case updates, evidence submissions, and hearing schedules through a unified interface, enhancing collaboration and transparency.

10.5 Blockchain for Evidence Immutable Ledger

While SHA256 hashing guarantees the integrity of individual files, implementing a private blockchain ledger for evidence movement (EvidenceChain) can provide an additional layer of tamper-proof security.

- **Immutable Evidence Tracking:** Every transaction, transfer, or verification of evidence can be recorded as a blockchain transaction. Once stored, records cannot be altered or deleted, providing an indelible audit trail for high-profile or sensitive cases.
- **Distributed Validation:** Blockchain ensures consensus across multiple nodes, making the system resilient against internal tampering or accidental data corruption. High-value cases, such as financial crimes or organized criminal investigations, benefit immensely from this level of security assurance.

- **Legal Credibility:** Courts increasingly recognize blockchain as a valid method for preserving digital evidence authenticity. Integrating this feature will strengthen CIMRS's legal reliability and future-proof the system against evolving forensic standards.

10.6 Advanced Forensic Module

The Evidence module can be expanded to provide comprehensive forensic capabilities, directly linking lab analyses and digital forensics into the case workflow.

- **Lab Management Sub-Module:** Forensic labs could update chemical, DNA, ballistic, or toxicology reports directly into CIMRS. This centralized approach reduces paperwork, eliminates delays, and ensures that investigation teams have immediate access to analytical results.
- **Digital Forensics:** Investigators often need to analyze mobile devices, hard drives, or networked systems. CIMRS could include specialized handling for forensic disk images and mobile dumps, ensuring automated verification of forensic image integrity via hashing. Integration with established forensic tools would streamline evidence analysis and reporting.
- **Automated Evidence Correlation:** The system could automatically correlate lab reports and digital findings with related case files, suggesting links or inconsistencies, thereby supporting investigators in building stronger cases.

11. References

11.1 Technical Reference Books

- **"Flask Web Development: Developing Web Applications with Python" by Miguel Grinberg:** This is the primary reference for the application's framework. It covers core concepts used in the project such as routing, Jinja2 templating, and the Flask-Login extension.
- **"Essential SQLAlchemy" by Jason Myers and Rick Copeland:** This book provides deep insights into the Object-Relational Mapping (ORM) techniques used to design the complex relationships between Cases, Criminals, and Evidence.
- **"Python CookBook" by David Beazley and Brian K. Jones:** Used as a reference for the file handling and cryptographic hashing (hashlib) logic implemented in the evidence module.
- **"Clean Code: A Handbook of Agile Software Craftsmanship" by Robert C. Martin:** Guided the modular implementation of the project, ensuring that the app.py logic is readable and maintainable.

11.2 Official Documentation (Technical Sites)

- **Flask Official Documentation (palletsprojects.com):** The definitive guide for implementing the app factory pattern and request handling.
- **SQLAlchemy Documentation (sqlalchemy.org):** Referenced for building the many-to-many relationship association tables (e.g., case_criminals).
- **Bootstrap 5 Documentation (getbootstrap.com):** Used for designing the responsive front-end components, navigation bars, and data tables.
- **Pandas Documentation (pandas.pydata.org):** Utilized for the implementation of the CSV/Excel export functionality in the Analyst module.
- **Python Hashlib Library:** Referenced for implementing the SHA256 integrity checks for digital evidence.

11.3 Domain & Legal Resources (Procedural Standards)

- **NIST Special Publication 800-86 (Guide to Integrating Forensic Techniques into Incident Response):** Provided the foundational logic for the Chain of Custody (EvidenceChain) and data integrity standards.
- **CCTNS (Crime and Criminal Tracking Network & Systems) Standards:** A reference for the standardized investigation stages (Preliminary Enquiry to Charge Sheet) used in the code's DEFAULT_INVESTIGATION_STAGES constant.
- **ISO/IEC 27037:2012:** Information technology — Security techniques — Guidelines for identification, collection, acquisition, and preservation of digital evidence.

11.4 Online Learning Platforms & Community Support

- **Stack Overflow:** A critical resource for debugging SQLAlchemy session management and Flask-Login integration issues.
- **GitHub Repositories:** Studied various Open Source Police Management Systems to understand best practices for RBAC (Role-Based Access Control) implementation.
- **Real Python (realpython.com):** Used for advanced tutorials on securing file uploads and implementing decorators for role verification.

11.5 Development Tools Used

- **Visual Studio Code (VS Code):** The primary Integrated Development Environment (IDE).
- **SQL Server for MySQL:** Used to manually verify the schema and data entries during the testing of the init-db command.
- **Postman:** Used to test the POST requests for investigation stage toggles and evidence transfers.

Index

Key Terms

A - Audit & Admin

- **ActivityLog (Model):** The database table responsible for storing every "write" operation for audit purposes.
- **Admin Role:** The highest privilege level; required for user management and accessing logs.
- **admin_required (Decorator):** A security wrapper that restricts access to administrative routes.
- **app_context:** The Flask mechanism used to handle database operations and CLI commands like init-db.

C - Case Management

- **Case (Model):** The central entity linking FIRs, evidence, criminals, and victims.
- **case_criminals (Table):** A many-to-many association table linking multiple suspects to multiple cases.
- **Case Investigation (Route):** The primary view for tracking the progress of an active investigation.
- **CourtCase (Model):** Stores judicial details like court names, case numbers, and trial statuses.

D - Data & Documents

- **db (SQLAlchemy):** The core object managing the application's connection to the MySQL database.
- **DEFAULT_INVESTIGATION_STAGES:** A predefined list of stages (e.g., "Preliminary Enquiry," "Charge Sheet") auto-assigned to every new case.
- **Document (Model):** Tracks case-related files such as legal reports, charge sheets, and versioning.

E - Evidence & Integrity

- **Evidence (Model):** Stores metadata for physical and digital evidence, including storage locations.
- **EvidenceChain (Model):** Manages the "Chain of Custody," recording every transfer of evidence.
- **Evidence Hashing:** The process of using **SHA256** to create a unique fingerprint for evidence files to prevent tampering.

F - Filing & Flow

- **FIR (Model):** The First Information Report table containing the initial details of a crime.
- **filed_at:** A timestamp field in the FIR model recording when a report was registered.
- **Flash Messages:** Temporary alerts used to confirm actions like "Case Registered" or "Logged Out."

H - Helpers & Hashing

- **hashlib:** The Python library used to compute cryptographic hashes for file integrity.
- **Hearing (Model):** Tracks trial schedules, including dates, hearing types, and judges.

L - Login & Logic

- **login_manager:** The Flask-Login extension used to handle user sessions.
- **log_action (Function):** A helper function that automatically inserts entries into the ActivityLog.

N - Notifications

- **Notification (Model):** Stores in-app alerts for users regarding case updates or tasks.

R - Roles & Security

- **ROLES (Constant):** A set defining the six authorized roles in the system (Admin, Officer, Investigator, etc.).

- **role_required (Decorator):** A custom function used to enforce Role-Based Access Control (RBAC) on specific routes.

S - Search & Schedules

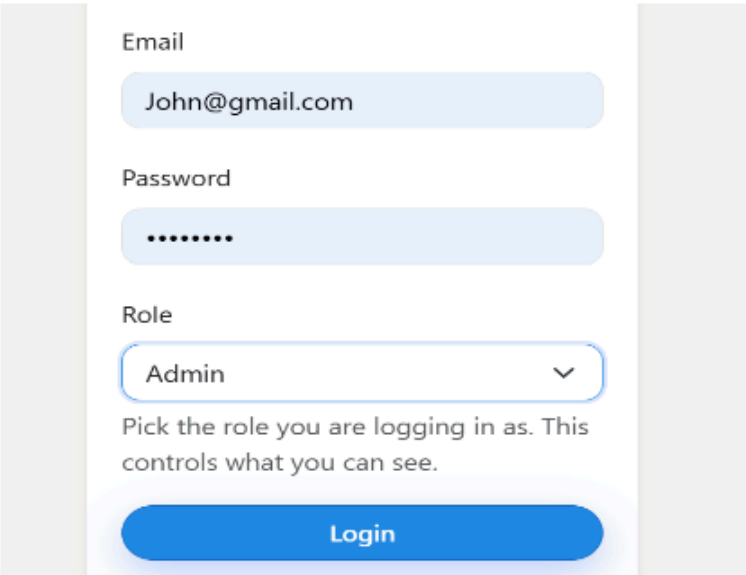
- **Schedule (Model):** Used for setting follow-up dates and trial milestones.
- **search (Global):** A comprehensive route using SQLAlchemy OR logic to search across all system entities.

U - Users

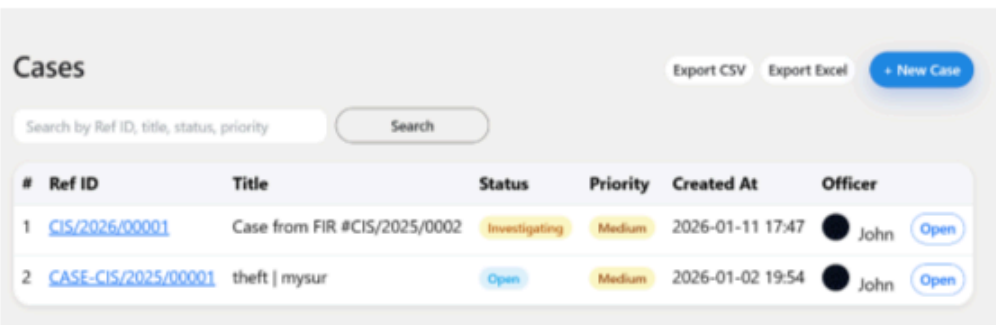
- **User (Model):** The core identity table storing names, emails, and PBKDF2:SHA256 hashed passwords.
- **UserMixin:** A class from Flask-Login providing standard implementations for user authentication methods.

12. Appendices

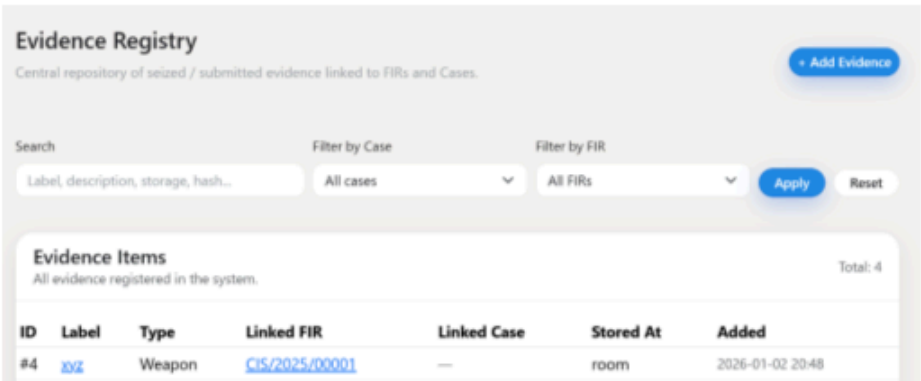
1. **Login Page** :



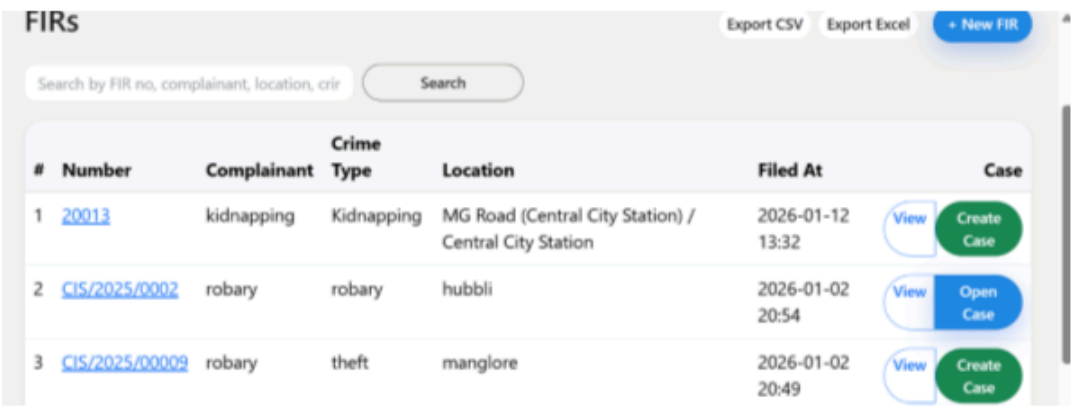
2. **Case Page :**



3 Evidenece Registry :



4. Fir Page :



12.1 About tools/technology

The system utilizes a specialized "Law Enforcement Tech Stack" chosen for its reliability and security.

- **Flask (Micro-framework):** Used for the backend logic. Its lightweight nature allows the system to run on standard police station hardware without heavy overhead.
- **SQLAlchemy (ORM):** Manages the complex relational mappings between cases, criminals, and victims. It prevents common SQL injection attacks through parameter binding.
- **Database Initialization Utility:** custom command-line utility is implemented to initialize the database schema and create default administrative accounts. This utility ensures consistent system setup, simplifies deployment, and reduces configuration errors during initial installation or reinitialization of the application environment.
- **Hashlib (SHA256):** Provides the cryptographic backbone for the system. SHA256 is an industry-standard algorithm that ensures even a single character change in a document or a single pixel change in a photo is detected.
- **Pandas (Data Analysis):** Used specifically for the "Analyst" role. It converts complex SQL joins into readable data frames for CSV and Excel exports.
- **Werkzeug (Security):** Handles PBKDF2 (Password-Based Key Derivation Function 2) hashing. This ensures that even if the database is stolen, user passwords remain computationally impossible to crack.
- Bootstrap 5: A frontend framework that ensures the system is "Mobile-First," allowing officers to use the system on smartphones or tablets while in the field.